Fixed labels

> Every anchored run so far drew its labels afresh on each visit to a context, so over a run the anchor saw labels on most `difference` contexts. Holding one draw fixed for the run, as a real labelled set would be, still keeps the anchor at every rate we tried. But the edit removes the op less reliably: a few runs end well below the rest, and more labels in the fixed set did not cure it.

The *post-hoc comparison plan* compares the edit along the anchored axis with directions found after training on a control. Writing it up, we noticed that the anchor labeller (inherited from M1) makes a fresh 1/50 draw on every visit to a context.

post-hoc comparison plan. Plans a comparison of the edit along e_1 with directions fitted after training on the controls at seeds 712 to 719, with the anchor's tolerance of scarce labels among the claims it tests.

An epoch covers about a quarter of the training split, so over 400 epochs a context is in view about a hundred times. With a fresh draw on each, about four fifths of the 42,891 `difference` contexts carry a label at some point (about 34,000), far more distinct labels than "one context in fifty" suggests. A real pipeline labels a fixed set once.

This report trains the recipe of record, `τ_ℓ fixed`, with one draw made per run and held for the whole run. We try three rates: 1/50 (about 870 labelled `difference` contexts), 1/25 (about 1,740), and 1/10 (about 4,320). Each rate runs at the eight seeds of the *fixed confirmation* (712 to 719). That experiment trained the same recipe with the per-visit draw, so its runs are the comparison.

fixed confirmation. `τ_ℓ fixed`, with the per-visit label draw, keeps the anchor on all eight runs at seeds 712 to 719 with removal near 0.9, and is the recipe of record for D2.2.

Observations

- The anchor settles at every rate (E1): all 24 runs are called kept at epoch 64, and α at the example answers settles about as fast as under the per-visit draw.

Alignment. How far a state points along e₁…

- Removal and spill at the end (E2): every run keeps the anchor, and runs pass the task gate as often as the per-visit runs do. At 1/50 and at 1/10, a few runs end well below the lowest per-visit run, so the seed mean falls. At 1/25 removal matches the per-visit draw, but one run spills late, over the criterion.

Removal. The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null.…

Spill. The largest drop on any other op, at the dose where it is largest.…

- When the low runs fall behind (E3): some low runs are behind from early in training, and others slide down after the anchor has settled. Neither kind tracks the label count.

Scope and method

This is an exploratory study, with no preregistration and no gate of its own. It started as a scout: one seed at 1/50, trained to epoch 64. Then 1/50 and 1/10 ran at four seeds to the end of training. At four seeds 1/50 removed less than 1/10, so they were carried to eight seeds, and 1/25 was added between them. The three rates were each chosen before their runs, and none was chosen by how it scored.

Every run is `τ_ℓ fixed` as the *fixed confirmation* trained it, with one change to the labels. In place of the per-visit draw, a single draw made before training picks the labelled contexts. A picked context carries the label on every visit, and any other context never does.

fixed confirmation. `τ_ℓ fixed`, with the per-visit label draw, keeps the anchor on all eight runs at seeds 712 to 719 with removal near 0.9, and is the recipe of record for D2.2.

Each seed has its own random stream for this draw, and the draws are nested, so every context labelled at 1/50 is also labelled at 1/25 and at 1/10. The per-visit draw is still taken and set aside, so each run trains on the same batches from the same initialization as the per-visit run at its seed. Only which contexts carry the pull differs.

The pull is a mean over the labelled contexts in each batch, so the label rate does not change how hard the anchor pulls. At 1/50 it also leaves the number of labelled contexts per batch unchanged on average. What the rate changes is how many distinct contexts the pull ever falls on, and how much the pull in a batch varies from one batch to the next.

We compare conditions as two samples of eight runs (the D2.2 rule since the determinism decision), with a 95% Welch t interval on the difference of seed means.[1] Over eight seeds, removal under the per-visit draw spreads by about 0.07.

1 Welch t interval…

Removal. The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null.…

All runs trained and were measured on L4 GPUs, as in the *fixed confirmation*. They share the controls of that experiment (no pull and no anti term, at the same seed).

The measurements

Every edit below is the e1 edit. Removal and spill are each a drop under that edit, less a baseline. The baseline is the median drop when the same edit is made along each of the other 63 coordinate axes of the run (the rule of the D2.2 design).

Removal. The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null....

Spill. The largest drop on any other op, at the dose where it is largest....

Removal

The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null. Higher is better, and a run below 0.5 has lost the anchor.

Spill

The largest drop on any other op, at the dose where it is largest. Lower is better, and a run above 0.02 is over the criterion.

Task change

The clean score per op, less that of the control at the seed. The gate of the D2.2 design asks for a mean change over ops of at least −0.02, and at least −0.05 on any one op.

Call

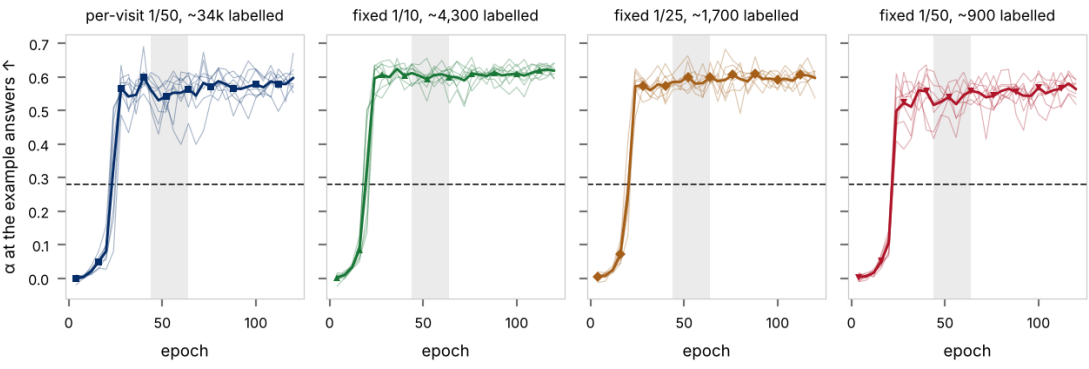
The early verdict on whether a run kept the anchor, from mean α at the example answers at slice 3 over epochs 44 to 64. A run below 0.15 is called lost, and one at 0.28 or above is called kept.

The anchor settles at every rate (E1)

With the pull confined to under a thousand contexts, does the anchor still settle on e_1 by the time the call is made? If a fixed set were too small, α at the example answers would rise more slowly or level off lower.

Call. The early verdict on whether a run kept the anchor, from mean α at the example answers at slice 3 over epochs 44 to 64....

Alignment. How far a state points along e_1...



The anchor settling, by condition. *α at the example answers of the `difference` probe contexts, at slice 3, along the first 120 epochs: the seed mean as a line with markers, each run faint behind it. The shaded band is the call window (epochs 44 to 64), and the dashed line is where the call counts a run as kept.*

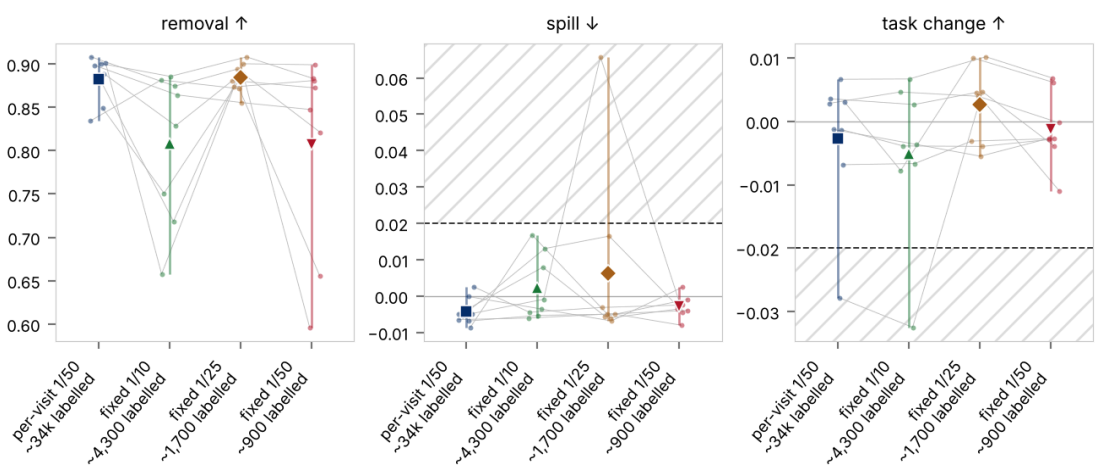
Every run is called kept, at every rate, with a wide margin. The rise over the first 40 epochs looks much the same in all four conditions. Over the call window, the seed mean of the fixed 1/50 runs is a little below that of the per-visit runs. The 1/25 and 1/10 runs are a little above it, and closer together. All of them are near the spread of the per-visit runs.

So a fixed set of under a thousand labelled contexts is enough for the anchor to settle. But across the fixed-label runs, mean α over the call window says little about how much a run removes at the end (Pearson $r = 0.04$ over the 24 runs).

Removal and spill at the end (E2)

Does a fixed labelled set remove `difference` as well as the per-visit draw, at the same cost to the other ops? If label count mattered over this range, removal would climb from 1/50 to 1/10.

Removal. The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null....



***Removal, spill, and task change at epoch 400, by condition**, from the most labelled `difference` contexts to the fewest. The seeds are small dots over a thin bar for their range, and the seed mean is the large mark; a grey line joins the runs of each seed, which share their initialization and batches. **Left:** removal at the full dose. **Middle:** spill, the largest net drop on another op at any dose; the dashed line is the criterion, with the side over it hatched. Both are net of the median over the other 63 axes of the run. **Right:** the task change, the mean clean score over ops less that of the control at the seed, with the gate of the D2.2 design dashed.*

labels	contexts labelled	removal ↑	removal less per-visit	spill ↓	over criterion ↓	over, block 1 ↓	task change ↑	task gate missed ↓
`per-visit 1/50`	34,000	0.88 (0.83 to 0.91)		−0.004 (−0.009 to 0.003)	0 of 8	0 of 8	−0.003 (−0.028 to 0.007)	1 of 8
`fixed 1/10`	4,324	0.81 (0.66 to 0.89)	−0.149 to −0.001	0.002 (−0.006 to 0.017)	0 of 8	0 of 8	−0.005 (−0.032 to 0.007)	1 of 8
`fixed 1/25`	1,742	0.88 (0.86 to 0.91)	−0.023 to 0.026	0.006 (−0.007 to 0.066)	1 of 8	1 of 8	0.003 (−0.005 to 0.010)	0 of 8
`fixed 1/50`	874	0.81 (0.60 to 0.90)	−0.173 to 0.021	−0.003 (−0.008 to 0.002)	0 of 8	0 of 8	−0.001 (−0.011 to 0.007)	0 of 8

***Removal, spill, and task change at epoch 400**, seed mean and range over eight runs at seeds 712 to 719. Removal and spill are net of the median drop over the other 63 coordinate axes of the run. "Contexts labelled" is the number of `difference` contexts that carried the label at some point in the run, the mean over seeds for a fixed draw and an estimate for the per-visit draw; both count over the whole corpus, of which training visits the first nine tenths. "Removal less per-visit" is a 95% Welch t interval on the difference of the seed means. "Over, block 1" counts the runs over the criterion under the same edit restricted to block 1. The task change is the mean over ops of the clean score less that of the control at the seed.*

Every run keeps the anchor, and the task change is near zero in every condition, with no more gate misses than under the per-visit draw. The differences are in removal, and in spill on one run.

Task change. The clean score per op, less that of the control at the seed....

Spill. The largest drop on any other op, at the dose where it is largest....

Removal does not climb with the label count. At 1/25 every run is within the per-visit range, and the seed means match.

At 1/50 and at 1/10 most runs remove about as much as the per-visit runs, but a few end below 0.8, under the lowest per-visit run (0.83): two at 1/50 (0.60 to 0.66) and three at 1/10 (0.66 to 0.75). Those runs pull the seed mean down by about the same amount at both rates.

The two rates have their low runs mostly at different seeds, and share only seed 719. At those seeds, the per-visit runs range from the lowest per-visit run (seed 715) to the second highest (seed 717; the grey lines in the figure). So the low runs look more like single runs that happened to end low than like seeds that are hard for the recipe.

All three low runs at 1/10 are at seeds 717 to 719, which is why the first look at four seeds missed them.

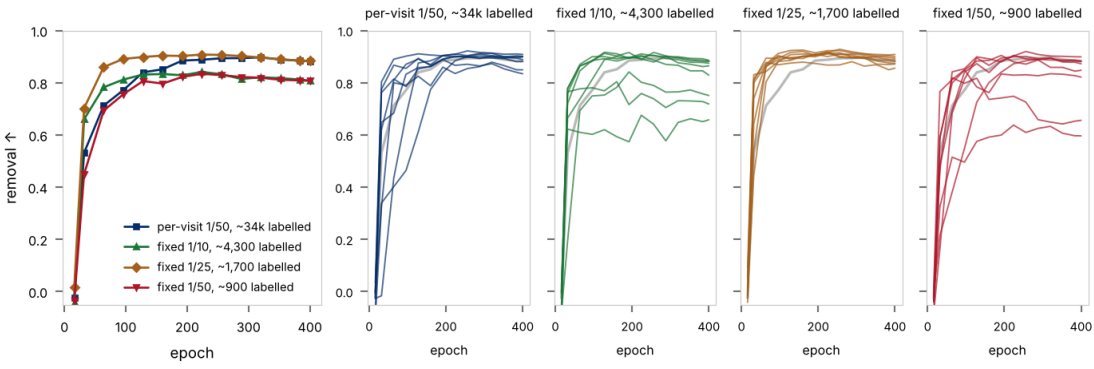
At 1/25, one run (seed 716) ends over the spill criterion, on `darken`. It is also over the criterion under the edit at block 1 alone. No other run in any condition goes over. Its spill rises late, from about epoch 192, like the late rise the *soft pool scout* saw on one seed of `τ_ℓ fixed`. Before that, it was over the criterion at only one checkpoint, just barely, at epoch 32 while the anchor was still settling.

soft pool scout. The pool over slices 1 to 3 with τ_ℓ held at 0.1 keeps the anchor on all eight seeds with no early spill, and one seed rises late to near the criterion.

The intervals on the removal difference from the per-visit draw are wide. At 1/10 the interval just excludes zero, and at 1/50 it does not. With eight runs each, a difference set by two or three low runs is hard to pin down. So we can't tell the three fixed rates apart, and 1/25 may look best only through the luck of its eight seeds.

When the low runs fall behind (E3)

Do the low runs fail to settle, or settle and then lose ground? The answer says whether the shortfall is about the anchor finding e_1 or about holding it there.



Removal along training, by condition. *Removal under the full edit at the checkpoints kept at epochs 16, 32 and 64 and then every 32 epochs, and at the end, net of the median over the other 63 axes of the run, as a share of the way from the clean score at the checkpoint to the target null at the end of the run.* ***Left:*** *the seed mean of each condition.* ***Right:*** *each run, one panel per condition, with the seed mean of the per-visit draw as a faint line behind for reference.*

Both happen. Some low runs are already behind at epoch 32 and never catch up, staying near 0.6 for the rest of training. Others come close to the per-visit level by epoch 100 or so and then slide down over the rest of the run, while the anti weight anneals. Under the per-visit draw, slow starters (such as seed 717, still near zero at epoch 32) catch up by the end.

Neither kind lines up with the rate. Both appear at 1/50 and at 1/10, and the 1/25 runs show neither. The 1/25 runs also reach their final level sooner than any other condition, by about epoch 64. With eight runs a condition, that may be the same luck of the seeds as in E2.

One account fits, though we have not tested it. With a fixed set, the pull keeps returning to the same few hundred or thousand contexts, so what the anchor learns depends on which contexts the draw happened to pick. A draw whose contexts cover the examples or the replacement noise unevenly could anchor the op only partly, and keep it that way.

The per-visit draw averages over almost every context, which would explain why its runs spread less. Telling this apart from plain run-to-run sensitivity would take several draws at the same seed.

Discussion

A fixed labelled set at one context in fifty is enough for the anchor to settle on e_1 and keep it (E1, E2). That was the main worry, since the per-visit draw had shown the anchor most of the `difference` contexts over a run. On that count, the scarce-label claim of the D2.2 design holds with labels as a real pipeline would have them.

But removal is less reliable with a fixed set: a few runs in eight end well short of the rest (E2), either from early in training or after a late slide (E3). Since a fixed set five times larger did no better, more labels of the same kind do not seem to be the fix, at least between about 900 and 4,300 contexts. The per-visit draw spreads less, and that may come from the variety of contexts it sees rather than their number.

Removal. The drop on `difference` under the full edit at every position and slice, as a share of the way from the clean score to the target null.…

If this holds, a comparison against post-hoc methods with fixed labels should expect a wider spread on the anchored side than the *fixed confirmation* showed. With that spread, eight runs give a noisy seed mean.

fixed confirmation. `τ_ℓ fixed`, with the per-visit label draw, keeps the anchor on all eight runs at seeds 712 to 719 with removal near 0.9, and is the recipe of record for D2.2.

References

fixed confirmation

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soft pool scout

: The pool over slices 1 to 3 with τ_ℓ held at 0.1 keeps the anchor on all eight seeds with no early spill, and one seed rises late to near the criterion.

post-hoc comparison plan

: Plans a comparison of the edit along e_1 with directions fitted after training on the controls at seeds 712 to 719, with the anchor's tolerance of scarce labels among the claims it tests.

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1. Welch t interval: a confidence interval for a difference of two means that does not assume the two samples have equal variance. ↩